

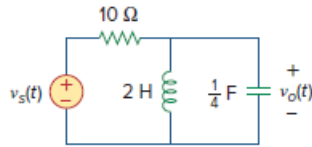
Problem

In the circuit shown in Fig. 17.69, the Fourier series expansion of $v_s(t)$ is

$$v_s(t) = 10 + \frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin(n\pi t)$$

Find $v_o(t)$.

Figure 17.69:



Step-by-step solution

Step 1 of 8

Refer to Figure 17.69 in the textbook.

Apply Kirchhoff's voltage law in the frequency domain.

$$\begin{aligned} \mathbf{I}_s &= \frac{1}{10 + \frac{(j\omega 2) \left(\frac{4}{j\omega} \right)}{j\omega 2 + \frac{4}{j\omega}}} \mathbf{V}_s \\ &= \frac{1}{10 + \frac{j\omega 8}{4 - 2\omega^2}} \mathbf{V}_s \\ &= \frac{1}{10 + \frac{j\omega 4}{2 - \omega^2}} \mathbf{V}_s \\ &= \frac{2 - \omega^2}{20 + j\omega 4 - 10\omega^2} \mathbf{V}_s \end{aligned}$$

Step 2 of 8

Consider the current through capacitor is, $\mathbf{I}_o$.

By current division, determine the expression for $\mathbf{I}_o$.

$$\begin{aligned} \mathbf{I}_o &= \frac{(j\omega 2)}{j\omega 2 + \frac{4}{j\omega}} \mathbf{I}_s \\ &= \frac{-\omega^2}{-\omega^2 + 2} \mathbf{I}_s \\ &= \frac{\omega^2}{\omega^2 - 2} \mathbf{I}_s \end{aligned}$$

Step 3 of 8

Substitute the expression for $\mathbf{I}_s$ in the equation.

$$\begin{aligned}\mathbf{I}_o &= \frac{\omega^2}{\omega^2 - 2} \cdot \frac{2 - \omega^2}{20 + j\omega 4 - 10\omega^2} \mathbf{V}_s \\ &= \frac{\omega^2}{10\omega^2 - j\omega 4 - 20} \mathbf{V}_s \\ &= \frac{\omega^2}{2(5\omega^2 - j\omega 2 - 10)} \mathbf{V}_s\end{aligned}$$

Step 4 of 8

Determine the expression for $\mathbf{V}_o$.

$$\begin{aligned}\mathbf{V}_o &= \mathbf{I}_o \mathbf{Z}_o \\ &= \frac{\omega^2}{2(5\omega^2 - j\omega 2 - 10)} \mathbf{V}_s \left(\frac{4}{j\omega} \right) \\ &= \frac{2\omega}{(5\omega^2 - j\omega 2 - 10)j} \mathbf{V}_s \\ &= \frac{2\omega}{5j\omega^2 + 2\omega - j10} \mathbf{V}_s \\ &= \frac{2\omega}{5j(\omega^2 - 2) + 2\omega} \mathbf{V}_s\end{aligned}$$

Step 5 of 8

Substitute $\omega_n = n\pi$ in the equation.

$$\begin{aligned}\mathbf{V}_o &= \frac{2\omega}{5j(\omega^2 - 2) + 2\omega} \mathbf{V}_s \\ &= \frac{2n\pi}{5j(n^2\pi^2 - 2) + 2n\pi} \mathbf{V}_s \\ &= \frac{2n\pi}{\sqrt{25(n^2\pi^2 - 2)^2 + 4n^2\pi^2} \angle \tan^{-1}\left(\frac{5(n^2\pi^2 - 2)}{2n\pi}\right)} \mathbf{V}_s \\ &= \frac{2n\pi}{\sqrt{25(n^2\pi^2 - 2)^2 + 4n^2\pi^2} \angle \tan^{-1}\left(\frac{5(n^2\pi^2 - 2)}{2n\pi}\right)} \mathbf{V}_s\end{aligned}$$

Step 6 of 8

For dc component, $\omega_n = 0$ or $n = 0$,

$$\begin{aligned}\mathbf{V}_{0,dc} &= \lim_{n \rightarrow 0} \left| \frac{2n\pi}{5j(n^2\pi^2 - 2) + 2n\pi} \times \mathbf{V}_{sdc} \right| \\ &= \lim_{n \rightarrow 0} \left| \frac{2n\pi}{5j(n^2\pi^2 - 2) + 2n\pi} \times 10 \right| \\ &= 0\end{aligned}$$

Step 7 of 8

For the n-th harmonic,

$$\begin{aligned} v_s(t) &= 10 + \frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin(n\pi t) \\ &= 10 + \frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \cos\left(n\pi t - \frac{\pi}{2}\right) \\ \mathbf{V}_s &= \frac{5}{n\pi} \angle -90^\circ \end{aligned}$$

Step 8 of 8

Determine the expression for $\mathbf{V}_o$.

$$\begin{aligned} \mathbf{V}_o &= \frac{2n\pi}{\sqrt{25(n^2\pi^2 - 2)^2 + 4n^2\pi^2}} \angle \tan^{-1}\left(\frac{5(n^2\pi^2 - 2)}{2n\pi}\right) \left(\frac{5}{n\pi} \angle -90^\circ\right) \\ &= \frac{10 \angle \left[-90^\circ - \tan^{-1}\left(\frac{5(n^2\pi^2 - 2)}{2n\pi}\right)\right]}{\sqrt{25(n^2\pi^2 - 2)^2 + 4n^2\pi^2}} \end{aligned}$$

Determine the time domain expression for $v_o(t)$.

$$\begin{aligned} v_o(t) &= 0 + \sum_{n=1}^{\infty} \frac{10 \cos\left[n\pi t - 90^\circ - \tan^{-1}\left(\frac{5(n^2\pi^2 - 2)}{2n\pi}\right)\right]}{\sqrt{25(n^2\pi^2 - 2)^2 + 4n^2\pi^2}} \text{ V} \\ &= \sum_{n=1}^{\infty} \frac{10 \sin\left[n\pi t - \tan^{-1}\left(\frac{5(n^2\pi^2 - 2)}{2n\pi}\right)\right]}{\sqrt{25(n^2\pi^2 - 2)^2 + 4n^2\pi^2}} \text{ V} \end{aligned}$$

Therefore, the time domain expression for $v_o(t)$ is,

$$v_o(t) = \sum_{n=1}^{\infty} \frac{10}{\sqrt{25(n^2\pi^2 - 2)^2 + 4n^2\pi^2}} \sin\left[n\pi t - \tan^{-1}\left(\frac{5(n^2\pi^2 - 2)}{2n\pi}\right)\right] \text{ V}.$$